GULF INTERNATIONAL BANK (GIB) INTERNAL AUDIT REPORT

ICAAP CREDIT RISK MODEL IMPLEMENTATION REVIEW

- **Bank:** Gulf International Bank B.S.C. (GIB)
- **Department:** Group Internal Audit – Risk Models & Quantitative Review Division
- **Date:** December 2026
- **Audit Cycle:** 2026 Review (First-Year Implementation Audit)

1. Executive Summary

Group Internal Audit has performed its inaugural review of the newly implemented Internal Capital Adequacy Assessment Process (ICAAP) Credit Risk Capital Model, deployed in 2026. This model constitutes a fundamental pillar of the Bank's internal economic capital assessment framework and is required to meet the stringent standards set forth under the **Central Bank of Bahrain (CBB) Rulebook Volume 1 (Module IC: ICAAP)** and **SAMA ICAAP Guidelines**.

Overall, the audit concludes that:

The model, as implemented and documented in 2026, does not currently meet the minimum standards of conceptual soundness, implementation quality, data governance, or documentation completeness expected for a robust Pillar 2 ICAAP framework.

Significant critical issues were identified across all operational dimensions:

- Incomplete and insufficient model documentation.
- Lack of clear, auditable end-to-end data lineage.
- Missing formal definitions for key macro and baseline risk inputs.
- No clear quantitative description of the simulation framework.
- Pervasive operational weaknesses and manual steps in the Excel implementation.
- Complete absence of model code version control.
- Missing empirical validation and stress-testing evidence.

- Material data quality issues within core portfolios.
- Ambiguity around data/model ownership roles and responsibilities.
- No clear traceability between source data, risk assumptions, methodology, and the final reported capital figures.

Despite receiving supplementary files, manual overrides, and partial access to sandbox systems, Group Internal Audit concludes that the model outputs are not fully reliable, and several critical economic capital assumptions remain unjustified or unsupported by empirical evidence.

2. Scope of Review

The 2026 audit covered:

- Model conceptual soundness and theoretical framework.
- Data source integrity and underlying data quality.
- Implementation pipelines (Excel macros, Python scripts, and process flows).
- Model documentation and user guides.
- Model governance, ownership, and sign-off workflows.
- Pillar 2 ICAAP reporting integration.
- Reproducibility and auditability of capital outputs.

Additional Materials Reviewed Beyond Formal Documentation

Due to severe gaps in the initial documentation package, Internal Audit had to request and manually reconstruct the following materials:

- Raw Probability of Default (PD) and Loss Given Default (LGD) historical data extracts (4 major CSV files).
- 17 Exposure at Default (EAD) tracking Excel workbooks (delivered as a partial subset due to systemic extraction delays).
- The underlying GCC corporate/retail segment exposure matrix used to compute final results.
- The Python simulation script `simulate.py` (obtained informally via direct developer email).
- System screenshots mapping the model's shared folder architecture.
- A draft backtesting template workbook (which was found to be largely unpopulated).

- Ad-hoc technical sessions with the risk modeling team to capture oral explanations of core model assumptions.

While such gaps can occur during a first-year implementation review, the severity of the documentation and pipeline deficiencies significantly exceeded Group Internal Audit's expectations.

3. Key Findings (Critical Issues)

Summary of total findings: 12 Critical, 7 Moderate, 4 Minor. The critical issues driving the negative audit opinion are detailed below.

Finding 1: Severe Documentation Insufficiency (Critical)

The 2026 model documentation package consists of only 6 baseline sections totaling approximately 3 pages of high-level text.

- **Key Missing Components:** Total absence of data source descriptions, formal methodology equations, asset segmentation logic, statistical calibration details, underlying risk correlation assumptions, validation results, and governance sign-offs. The only annex provided was a basic, 4-line variable list.
- **Audit Conclusion:** The current documentation fails to meet any minimum baseline required by CBB Module IC or SAMA Pillar 2 guidelines.
- **Recommendation 1:** Author and deliver a comprehensive Model Development Documentation manual (expected length: 15–25 pages) explicitly detailed to cover the complete lifecycle of data lineage, methodology, validation, and governance.

Finding 2: No Data Lineage / Traceability (Critical)

No systematic information or map is provided regarding source core-banking platforms, file ingestion structures, extraction, transformation, and loading (ETL) routines, data stewardship/ownership, quality control (QC) checks, or the creation of intermediate modeling datasets. Internal Audit was forced to manually map the data lineage through employee interviews.

- **Recommendation 2:** Establish and document an end-to-end automated data lineage map showing every step from initial system extraction to final model capital calculations.

Finding 3: Unexplained PD and LGD Inputs (Critical)

The model documentation provides zero reconciliation protocols for underlying PD and LGD parameters.

- **Audit Findings:** Internal Audit discovered that manual PD overrides within the internal credit rating system were omitted from the modeling files. LGD inputs were highly inconsistent across sheets (showing unexplained variances of 1 to 5 percentage points), multiple missing PD records had no documented imputation method, and LGDs sourced from the Recoveries department lacked any economic downturn adjustments.
- **Recommendation 3:** Formalize an automated PD/LGD extraction, cleaning, and reconciliation framework embedded with strict data completeness and quality control rules.

Finding 4: Methodology Not Documented (Critical)

The documentation vaguely states that *"A simulation-based approach is used."* However, the review revealed:

- No explanation of the target loss distribution, simulation counts, portfolio correlations, or mathematical loss formulas.
- The emailed Python script (`simulate.py`) revealed undocumented, ad-hoc lognormal distribution assumptions, a simulation count of only 5,000 (contradicting verbal claims of 20,000), a hardcoded correlation parameter set arbitrarily at 0.18, no fixed seed control (preventing exact calculation replication), and no input exception handling.
- **Recommendation 4:** Formally document and lock the model methodology, including all governing statistical equations, distributional choices, parameter calibrations, and structured simulation seed controls.

Finding 5: Weak and Error-Prone Implementation (Critical)

The technical execution of the model relies on a highly unstable, unindustrialized computing environment:

- **Excel Environment:** 17 distinct EAD files are merged manually by analysts; broken formulas (`#REF!`, `#VALUE!`) were discovered across 3 active sheets; hidden columns contain conflicting exposure metrics; and live data links point to the local C-drives of former employees. Cells and sheets lack write-protection.

- **Python Environment:** Script contains hardcoded local folder paths, zero logging capabilities, and automatically overwrites prior run outputs without backup. The code is not checked into an institutional version control system (e.g., Git).
- **Recommendation 5:** Industrialize the entire modeling pipeline. Eliminate manual Excel aggregation steps, remove hardcoded configurations, and move the Python code into a secure, version-controlled institutional repository.

Finding 6: Unreliable Data Quality (Critical)

Data scanning by the audit team revealed significant anomalies across the modeling portfolio that were completely omitted from risk reporting:

- 2.4% of active counterparties lacked assigned PDs.
- 17.0% of historical default files lacked recorded recovery values.
- 1.1% of records exhibited negative EAD values.
- Widespread systemic misclassification of SME flags and duplicated counterparty lines.
- Inverted and broken cross-joins between the primary LGD and PD lookup matrices.
- **Recommendation 6:** Define and deploy an institutional data quality framework for the ICAAP model, featuring strict validation thresholds, exception alerts, and clear remediation workflows.

Finding 7: No Validation or Testing Evidence (Critical)

The 2026 documentation features zero backtesting, sensitivity metrics, or model stability analysis. When pushed for evidence, the modeling team provided a partially empty Excel worksheet (`Backtest.xlsx`) that completely lacked Retail portfolio data, and an informal email stating that *"sensitivity is broadly stable."* This does not satisfy CBB regulatory standards.

- **Recommendation 7:** Task an independent function to execute and document a comprehensive initial model validation cycle, encompassing backtesting, parameter sensitivity, portfolio stability, and macro-scenario consistency tests.

Finding 8: Questionable Model Output Reliability (Critical)

The primary economic capital output submitted for ICAAP compliance was **$1.68 billion**. Internal Audit attempted to replicate this figure and generated highly disparate results across environments:

- *Audit Team Independent Recalculation:* **$1.74 billion**
- *Raw Python Code Execution:* **$1.63 billion**
- *Intermediate Excel Sheet Calculation:* **$1.71 billion**

Unexplained variations ranging from **$90 million to $110 million** exist between these environments with no clear reconciliation trail.

- **Recommendation 8:** Establish a single, definitive corporate source of truth for model execution to ensure absolute data reproducibility and eliminate cross-platform output variances.

Finding 9: Governance Not Operational (Critical)

The model has been utilized without formal institutional guardrails. There are no signed model approval documents, no independent technical validation opinions, no model risk committee sign-offs, and no formal change/version log tracking adjustments to the code or assumptions.

- **Recommendation 9:** Integrate the model into the formal GIB Model Risk Management policy framework, ensuring distinct lines of ownership, formal committee review, annual sign-offs, and an immutable change log.

Finding 10: Weak ICAAP Integration (Critical)

The final ICAAP submission document directly pulls numbers from this model but fails to provide any explanatory narrative, qualitative commentary, or bridge analysis reconciling these internal economic capital metrics with regulatory Pillar 1 capital. Furthermore, there is no operational connection between model outputs and the Bank’s Risk Appetite Framework (RAF).

- **Recommendation 10:** Align the ICAAP narrative with the quantitative outputs of the model, building an explicit analytical link between internal capital models, forward-looking stress scenarios, and Board-approved risk appetite limits.

4. Moderate and Minor Findings

The core drivers of the negative assessment are detailed above; less critical findings are summarized below for inclusion in the report Annex.

- **Moderate Issues:** Missing segmentation rationale for specialized lending portfolios; complete absence of geographical concentration mapping across the GCC; lack of explanatory commentary regarding a significant PD spike observed in 2024 data; undefined baseline and adverse stress scenario generation logic; and unarchived, loose historical modeling files.
- **Minor Issues:** Minor cosmetic and formatting errors across reporting dashboards, inconsistent case syntax in code variable names, and non-standard file naming conventions across shared network drives.

5. Overall Conclusion

The ICAAP Credit Risk Capital Model, as implemented for the 2026 reporting cycle, **is not fit for purpose in its current form.** While the development team's initiative represents a foundational first step toward building an advanced internal capital assessment framework, substantial corrective engineering and documentation are required before Group Internal Audit can deem the model compliant with internal policies, SAMA directives, or CBB regulatory expectations.

6. Summary of Recommendations (High-Level Actions)

1. **Rewrite Documentation:** Author an end-to-end Model Development Document.
2. **Reconstruct Data Lineage:** Map and automate the flow from source systems to the model.
3. **Parameter Reconciliation:** Standardize and clean the PD/LGD data extraction processes.
4. **Methodology Formulation:** Fully document mathematical equations, distributions, and seeds.
5. **Pipeline Industrialization:** Eliminate manual Excel data manipulation and implement Git version control.
6. **Data Quality Controls:** Establish automated thresholds and exception data checks.

7. **Independent Validation:** Complete comprehensive backtesting and sensitivity analysis.
8. **Ensure Reproducibility:** Resolve code-to-spreadsheet discrepancy variances to establish a single source of truth.
9. **Enforce Governance:** Secure official Model Risk Committee approvals and establish clear ownership lines.
10. **Pillar 2 Alignment:** Integrate the model's outputs with the broader ICAAP narrative and the Risk Appetite Framework.